

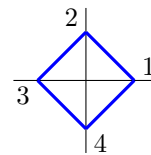
Consider a square placed on Cartesian plane such that the square's center is at the origin and its corners are on the axes (like a diamond; see the figure below). Any motion of the square which keeps the center at the origin and corners on the axes has the same result as one of eight motions: $r_0 \equiv$ rotation of 0° , $r_1 \equiv$ rotation of 90° , $r_2 \equiv$ rotation of 180° , $r_3 \equiv$ rotation of 270° , $s_0 \equiv$ x -axis reflection, $s_1 \equiv$ reflection on $y = x$, $s_2 \equiv$ y -axis reflection, or, and $s_3 \equiv$ reflection on $y = -x$.

There are exactly 8 unique ways to position the square. There are four places to position corner 1. Corner 2 must be adjacent to 1; so there are two choices for its place giving 8 possibilities. Corners 3 and 4's placement is then determined as they must be adjacent to 2 and 1 respectively.

Theorem. $D_4 = \{r_0, r_1, r_2, r_3, s_0, s_1, s_2, s_3\}$ is a group under composition of these motions. The set D_4 is called the dihedral group of degree 4 (the group of symmetries of a regular 4 sided polygon). First

Proof. The identity is r_0 . (The inverse of a rotation, r_a , is the rotation, r_b , such that $a + b = 0 \pmod{4}$, i.e., adds up to 2π . Also $s_n^{-1} = s_n$) For associativity, note that repositioning the corners is a bijective function from $f : \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4\}$, i.e., it is a permutation on $\{1, 2, 3, 4\}$. So each motion is a function, and function composition is associative. For closure and inverses, see the Cayley table below:

	r_0	r_1	r_2	r_3	s_0	s_1	s_2	s_3	s_4
r_0	r_0	r_1	r_2	r_3	s_0	s_1	s_2	s_3	s_4
r_1	r_1								
r_2	r_2								
r_3	r_3								
s_0	s_0								
s_1	s_1								
s_2	s_2								
s_3	s_3								



□

Alternatively,

Proof. The motions are linear transformations. So each motion can be represented with a 2×2 matrix.

$$r_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, r_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, r_2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, r_3 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$s_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, s_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, s_2 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, s_3 = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$$

Where a corner's Cartesian coordinates (x, y) are moved via matrix multiplication:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}$$

$$r_0 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}, r_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}, r_2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ -y \end{pmatrix}, r_3 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ -x \end{pmatrix}$$

$$s_0 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ -y \end{pmatrix}, s_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}, s_2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ y \end{pmatrix}, s_3 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ -x \end{pmatrix}$$

r_0 is the identity. Matrices are associative. Each matrix is square, and has a non-zero determinant. So it is easy to check with $\frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1}$ that each element has an inverse in D_4 . For closure, recall that the $\det(A)\det(B) = \det(AB)$. The determinant for each element in D_4 is ± 1 . So the determinant of the product of any two elements in D_4 will also be ± 1 . Since the only entries in the matrices are 0 and 1, $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$ implies that $a = d = 0$ or $b = c = 0$ with the remaining values being ± 1 . As all 8 such possibilities are in D_4 this implies that the product must also be in D_4 . □